Risk Management and Financial Institutions

Mock exam

****

You have 60 minutes time.
Maximum score: 80.
A minimum of 50% is to be achieved to pass the exam. Your final grade depends on the scores you gathered during the semester.

****

1) Suppose that a portfolio consists of two shares as follows:

	Share A	Share B
Value of a share	30 $	20$
Number of shares	10,000	30,000
Expected return (annual)	14%	20%
Volatility (annual)	20%	30%

The correlation coefficient of the returns is 0.5. The portfolio's return is assumed to be normally distributed.

a) What is the volatility and expected return of the portfolio for 6 month?
b) Calculate the six-month 95% VaR and ES of the portfolio!
c) You think, the correlation of the returns will increase in the future. How does your expectations effect the risk measures?

(10 points)

2) What are the main assumptions of Merton model? Explain the terms distance to default and expected default frequency in this model. (8 points)

3) What is the definition of crisis. What are the operational answers for a crisis? (8 points)

4) Suppose that a 5-year zero-coupon Treasury bond with a face value of 100 yields 3% and a similar 5-year zero-coupon bond issued by a corporation yields 4%. (Both rates are logreturns.) Make the simplifying assumption that there are no recoveries in the event of a default. Calculate the default probability of the corporation! How to interpret the PD you calculated? (8 points)

5) Suppose each of two independent projects has a probability of 4% of a loss of $5 million and a probability of 96% of a loss of $1 million during a one-year period. Calculate the one-year 95% VaR and ES of each portfolio! Calculate the one-year 95% VaR and ES of a portfolio containing both projects!
Compare the two risk measures according to the results (12 points)

6) In a currency swap, interest on a principal in one currency is exchanged for interest on a principal in another currency. The principals in the two currencies are exchanged at the end of the life of the swap. Why is the credit risk on a currency swap greater than that on an interest rate swap? (8 points)

7) How do you define trading book? What are the elements of capital requirements based on the trading book positions? (10 points)

8) Explain why a new transaction by a bank with a counterparty can have the effect of increasing or reducing the bank's credit exposure to the counterparty. (6 points)

9) Describe the new proposals of Basel 3 regulation! (10 points)
Short explanation is required!